DPP	DEPARTMENT POLICY AND PROCEDURE		
Title	**FUNDAMENTALS OF CRITICAL CARE SUPPORT**		
Department	**Pediatric Intensive Care Unit**		
Applies to	**All PICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-PICU-013 (v3)**		
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1.0 PURPOSE

1.1. To unify the nursing care to pediatric patients develop life threatening arrhythmias.

1.2. To control life threatening arrhythmias and prevent recurrence.

1.3. To prevent injury and complication.

2.0 DEFINITION/ABBREVIATION

2.1. **CPR** – Cardio Pulmonary Resuscitation

2.2. **ECG** – Electro Cardio Gram

2.3. **PEA** – Pulseless electrical activity

2.4. **AED** – Automatic external defibrillator.

2.5. **Pediatric Fundamental Critical Care Support (PFCCS) -** Pediatric Fundamental Critical Care Support is a major resource for those learning how to recognize critical illness and initiate care for the critically ill pediatric patient.

2.6. **Bradycardia -** Bradycardia is defined as a heart rate below the normal range for age

2.7. **Tachycardia -** Tachycardia is a condition in which the heart rate is abnormally high.

2.8. **Pulseless electrical activity (PEA) -** Pulseless electrical activity is a clinical condition characterized by unresponsiveness and lack of palpable pulse in the presence of organized cardiac electrical activity

2.9. **Ventricular tachycardia -** Ventricular tachycardia is a type of tachycardia, or a rapid heartbeat that arises from improper electrical activity of the heart presenting as a rapid heart rhythm that starts in the bottom chambers of the heart called the ventricles.

2.10. **Ventricular fibrillation -** Ventricular fibrillation is the most serious cardiac rhythm disturbance. The lower chambers quiver and the heart can't pump any blood, causing cardiac arrest.

3.0 RESPONSIBILITY

3.1 **Physician** is responsible in ordering medications and possible procedures.

3.2 **Head Nurse** must ensure that all antiarrhythmic drugs including controlled drugs are available in the unit.

3.3 **Charge Nurses** is responsible to assign a competent nurse to patient having arrhythmias and to double check orders including medications.

3.4 **Staff nurses** are responsible in assessing and re-assessing patient throughout to document the whole event and carry out the care needed.

4.0 POLICY

4.1. Basic life support measures (BLS) shall be initiated immediately for any pediatric patient who experiences acute cardiopulmonary distress.

4.2. Code Team members function collaboratively during the code with one person identified as the code team leader.

4.3. PICU staff will assist the Code Blue team as directed.

4.4. The individual assigned as recorder will document all treatments, medications, ECG data etc. on the Code Blue record.

4.5. Code Carts are maintained in specified areas throughout the hospital (as designated in the code blue manual), and will be checked daily during business hours.

4.6. A Pediatric Code Blue should be initiated for any acute physiological disturbance requiring immediate attention.

5.0 PROCEDURE (S)

5.1. ***Preparatory phase***

5.1.1. Understands criteria for the initiation of CPR.

5.1.2. Recognize and identify the risk of deterioration in patient's vital signs and conscious status.

5.1.3. Early identify the rhythm causing cardiorespiratory compromise.

5.2. ***Performance Phase***

5.2.1. **PAEDIATRIC BRADYCARDIA**

5.2.1.1. Assess and support airway, breathing and circulation as needed.

5.2.1.2. Give 100% oxygenation. To maintains oxygen saturation.

5.2.1.3. Attach monitor or defibrillator when available.

5.2.1.4. Assess bradycardia still persisting or not and assess the severity due to it.

5.2.1.5. **If not persisting:**

5.2.1.5.1. Support airway, breathing and circulation and give oxygen.

5.2.1.5.2. Observe the patient closely

5.2.1.5.3. Consider expert consultation Provides supportive management for patient.

5.2.1.6. **If still persisting bradycardia and which causes cardiorespiratory compromise,** Perform CPR

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5.2.1.7. **If persistent symptomatic bradycardia**

5.2.1.7.1. Give epinephrine via:

5.2.1.7.1.1. Intravenous or intraosseous route 0.01mL/kg (1: 10000 = 0.1mL/kg)

5.2.1.7.1.2. Endotracheal tube 0.1mL/kg (1:1000 = 0.1mL/kg) and repeat 3 to 5 minutes.

5.2.1.8. If increased vagal tone or primary AV block: Give atropine, first dose: 0.02mg/kg, may repeat. (Minimum dose: 0.1mg; maximum total dose for child: 1mg).

5.2.1.9. Consider cardiac pacing

5.2.1.10. **If pulseless arrest develops,** manage as per pulseless arrest (Refer pulseless arrest unit competency).

5.2.2. **PAEDIATRIC TACHYCARDIA**

5.2.2.1. Support airway, breathing and circulation as needed.

5.2.2.2. Give 100% oxygenation. To know the health status of patient.

5.2.2.3. Attach monitor or defibrillator when available.

5.2.2.4. Evaluate QRS duration in ECG, if still symptoms persist.

5.2.2.5. **If wide QRS (>0.08s)** - Check for possible ventricular tachycardia.

5.2.2.6. **If narrow QRS (<0.08s)** - Evaluate rhythm with 12-lead ECG or monitor in order to identify possible sinus tachycardia or probable supraventricular tachycardia.

5.2.2.7. **Management of probable sinus tachycardia**

5.2.2.7.1. Identify probable sinus tachycardia with ECG shows normal P waves, Variable R-R interval, constant P-R interval.

5.2.2.7.2. Check history, which is consistent with known cause.

5.2.2.7.3. Infants rate usually < 220bpm and children <180bpm.

5.2.2.7.4. Treat the cause.

5.2.2.8. **Management of supraventricular tachycardia:**

5.2.2.8.1. Identify the history compatible with supraventricular tachycardia.

5.2.2.8.2. Identify probable sinus tachycardia with ECG shows P waves absent or abnormal, heart rate not variable, Variable R-R interval, and constant P-R interval.

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5.2.2.8.3. Check history, which is consistent with known cause.

5.2.2.8.4. Infants rate usually > 220bpm and children180bpm.

5.2.2.8.5. Consider vagal maneuvers.

5.2.2.8.6. If IV access readily available, give adenosine 0.1mg/kg (maximum 6mg) by rapid bolus.

5.2.2.8.7. Give synchronized cardio version: 0.5 to1J/kg, if not effective, increase to 2J/kg. Sedate if possible but don't delay cardioversion.

5.2.2.8.8. If still persists, Consider amiodarone 5mg kg Intravenous over 20-60 minutes or procainamide 15mg/kg Intravenous over30 - 60 minutes. Do not administer these both drugs together.

5.2.2.9. **Management of possible ventricular tachycardia:**

5.2.2.9.1. Give synchronized cardio version: 0.5 to 1J/kg, if not effective, increase to 2J/kg. Sedate if possible but don't delay cardio version.

5.2.2.9.2. Give adenosine 0.1mg/kg (maximum 6mg) by rapid bolus.

5.2.2.9.3. Consider amiodarone 5mmg/ kg Intravenous over 20-60 minutes or procainamide 15mg/kg Intravenous over 30 -60 minutes. Do not administer these both drugs together.

5.2.2.9.4. Search for and treat possible contributing factors. To identify etiological factors.

5.2.3. **PULSELESS ARRHYTHMIAS**

5.2.3.1. Begin CPR while preparing equipment. To act as needed.

5.2.3.2. Give 100% oxygenation. To know the health status of patient.

5.2.3.3. Attach monitor or defibrillator when available

5.2.3.4. Evaluate Rhythm and identify whether it is shock able or not shock able

5.2.3.5. **Non shockable – asystole/ Pulseless electrical activity.**

5.2.3.5.1. Resume CPR immediately. To resuscitate patient.

5.2.3.5.2. Give epinephrine via intravenous or intraosseous route 0.01mL/kg (1: 10 000: 0.1mL/kg) and via endotracheal tube 0.1mL/kg (1: 1000:0.1mL/kg) and repeat 3 to 5 minutes.

5.2.3.5.3. Check rhythm whether it is shockable or not

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5.2.3.5.4. If not shockable:

5.2.3.5.4.1. If a systole manages with drugs and CPR.

5.2.3.5.4.2. Do not give shock.

5.2.3.5.4.3. If electrical activity, check pulse, if no pulse, manage as asystole.

5.2.3.5.4.4. If pulse present, begin post resuscitation care.

5.2.3.6. **Non shockable – asystole/ Pulse less electrical activity.**

5.2.3.6.1. Resume CPR immediately. To resuscitate patient effectively.

5.2.3.6.2. Give epinephrine via intravenous or intraosseous route 0.01mL/kg (1: 10 000: 0.1mL/kg) and via endotracheal tube 0.1mL/kg (1: 1000:0.1mL/kg) and repeat 3 to 5 minutes.

5.2.3.6.3. Check rhythm whether it is shock able or not.

5.2.3.6.4. If not shockable:

5.2.3.6.4.1. If a systole manages with drugs and CPR. Do not give shock.

5.2.3.6.4.2. If electrical activity, check pulse, if no pulse, manage as asystole.

5.2.3.6.4.3. If pulse present, begin post resuscitation care.

5.2.3.7. **Shockable - Ventricular Fibrillation / Pulseless Ventricular Tachycardia.**

5.2.3.7.1. Give shock manually or using AED: Defibrillate 2J/kg monophasic (or biphasic equivalent). Resume CPR immediately. Administer CPR 5 cycles / 2 minutes

5.2.3.7.2. Check rhythm whether it is shockable or not.

5.2.3.7.3. If not shockable: manage as above mentioned

5.2.3.7.4. If shockable: Give shock for persistent Ventricular fibrillation and pulseless ventricular tachycardia. Defibrillate 4 J/kg monophasic (or biphasic equivalent). Resume CPR immediately. Give epinephrine via intravenous or intraosseous route 0.01mL/kg (1: 10,000 = 0.1mL/kg) and via endotracheal tube 0.1mL/kg (1:1000 = 0.1mL/kg) and repeat 3 to 5 minutes.

5.2.3.7.5. Resume CPR immediately. Administer CPR 5 cycles / 2 minutes.

5.2.3.7.6. Check rhythm whether it is shockable or not.

5.2.3.7.7. If not shockable, manage as above mentioned.

5.2.3.7.8. If shockable: Give shock for persistent Ventricular fibrillation and pulseless ventricular tachycardia. Defibrillate 4 J/kg monophasic (or biphasic equivalent).

5.2.3.7.9. Resume CPR immediately. To resuscitate patient effectively.

5.2.3.7.10. Consider antiarrhythmics (e.g. Amiodarone 5mg/kg via intravenous or intraosseous route or lidocaine 1mg/kg via intravenous or intraosseous route). Consider magnesium 25 to 50 mg/kg via intravenous or intraosseous route, maximum 2g for torsades de pointes

5.2.3.7.11. Resume CPR immediately. Administer CPR 5 cycles / 2 minutes

5.2.3.7.12. Avoid hyper ventilation.

5.2.3.7.13. Search for and treat possible contributing factors.

5.2.3.7.14. Secure airway and confirm placement.

5.2.3.7.15. After an advanced airway is placed, rescuers no longer delivery "cycles" of CPR. Give continuous chest compressions without pauses or breaths. Give 8 to 10 breaths/ minute. Check rhythm every 2 minutes. Improves airway, circulation and breathing.

5.2.3.7.16. Resume CPR immediately. Administer CPR 5 cycles / 2 minutes

5.3. ***Follow-up phase***

5.3.1. Document procedure with time and after effect of the action.

5.3.2. Evaluate patient in order to detect early in deterioration of condition.

6.0 EQUIPMENTS

6.1. Crash cart / resuscitation equipment

7.0 ATTACHMENT/FORMS

7.1. N/A

8.0 REFERENCES

8.1. MOH Guidelines for Pediatric Care

8.2. DPP-PICU-0026(V2) – Fundamentals of Critical Care Support – PMSH, 01/10/1441

8.3. Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) PICU standards, 3rd version, January 2016

9.0 REVISION CLAUSE

9.1. Used the new logo for AL-Baha Health Cluster.

9.2. Modify the policy and procedure lay out, changed the content of the Header and footer

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